

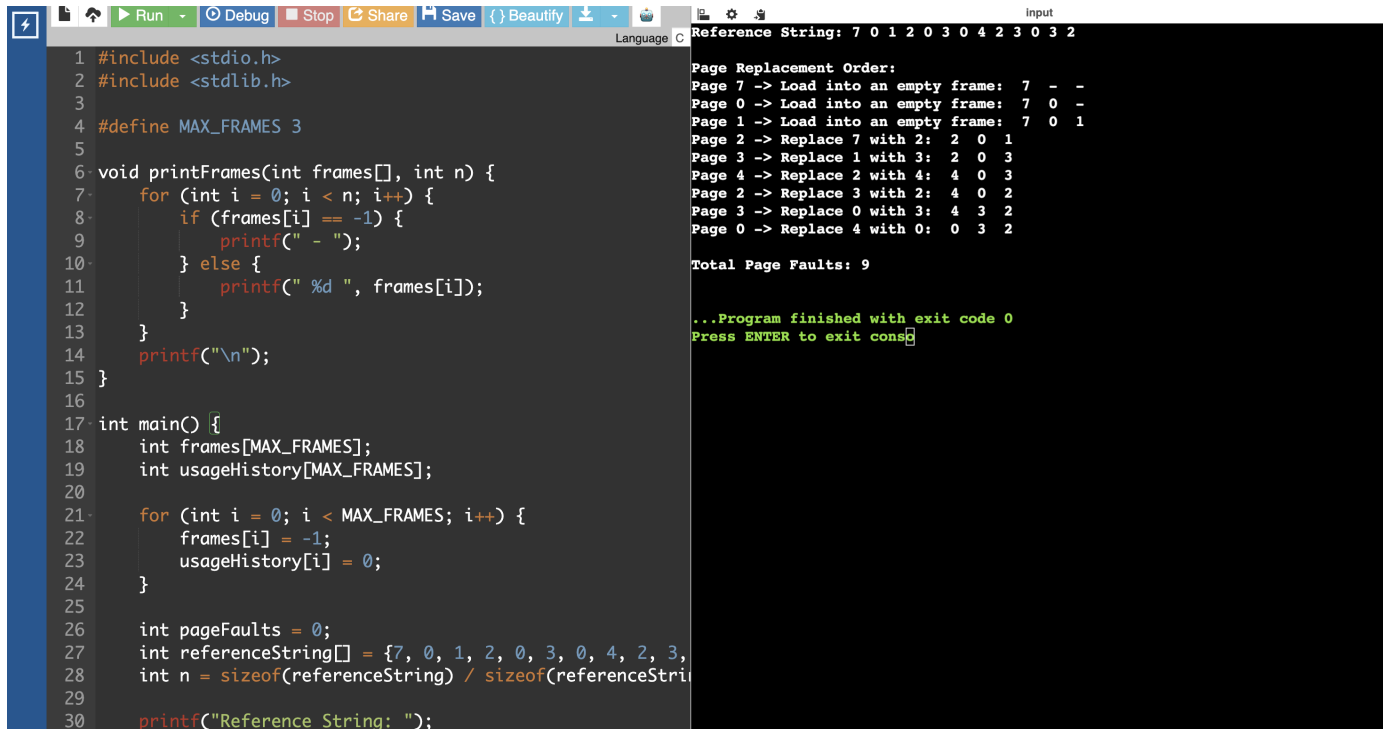
CSA0404 – OPERATING SYSTEMS

EXPERIMENT 32

QUESTION

Construct a C program to simulate the Least Recently Used paging technique of memory management.

PROGRAM



```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 #define MAX_FRAMES 3
5
6 void printFrames(int frames[], int n) {
7     for (int i = 0; i < n; i++) {
8         if (frames[i] == -1) {
9             printf(" - ");
10        } else {
11            printf(" %d ", frames[i]);
12        }
13    }
14    printf("\n");
15 }
16
17 int main() {
18     int frames[MAX_FRAMES];
19     int usageHistory[MAX_FRAMES];
20
21     for (int i = 0; i < MAX_FRAMES; i++) {
22         frames[i] = -1;
23         usageHistory[i] = 0;
24     }
25
26     int pageFaults = 0;
27     int referenceString[] = {7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0};
28     int n = sizeof(referenceString) / sizeof(referenceString[0]);
29
30     printf("Reference String: ");
```

Reference String: 7 0 1 2 0 3 0 4 2 3 0 2

Page Replacement Order:

Page 7 -> Load into an empty frame: 7 - -

Page 0 -> Load into an empty frame: 7 0 -

Page 1 -> Load into an empty frame: 7 0 1

Page 2 -> Replace 7 with 2: 2 0 1

Page 3 -> Replace 1 with 3: 2 0 3

Page 4 -> Replace 2 with 4: 4 0 3

Page 2 -> Replace 3 with 2: 4 0 2

Page 3 -> Replace 0 with 3: 4 3 2

Page 0 -> Replace 4 with 0: 0 3 2

Total Page Faults: 9

...Program finished with exit code 0

Press ENTER to exit console